

# Nash Bargaining for Wholesaler Territory Division

## A discrete formulation over postal codes

### Abstract

Two merging wholesaling forces must divide a common territory. We formulate the problem directly over postal codes and solve it by Nash bargaining from the zero disagreement point: neither wholesaler can unilaterally revert to a pre-merger book once the merger is executed, so the outside option is nothing, not the legacy territory. Ordering zips by the ratio of the two wholesalers' utilities gives a fast heuristic and, via outer approximation on the concave log-objective, an exact solution with an optimality certificate. Nash is Pareto efficient by construction, satisfies envy-freeness up to one zip, and cannot purchase fairness by making both parties worse off — a failure mode the equalisation criteria do exhibit. Contiguity is then imposed exactly by a districting integer program — solved by outer approximation, since the log of the Nash product is concave rather than linear — and costs under half a percent of the bargaining product. A fifty-zip instance is worked through in full: allocation, contiguity, contestability triage, and exact parameter breakpoints. Alternative solution concepts and the constructions specific to them are collected in the appendices.

## 1 Setup

Index postal codes  $z = 1, \dots, n$ , each carrying pre-merger sales  $A_z$  to wholesaler  $a$  and  $B_z$  to wholesaler  $b$ , and total market opportunity  $M_z$  (demand net of what either has already written); write  $M = \sum_z M_z$ . Two parameters govern how a zip's value splits between whichever wholesaler is assigned it and whichever is not. *Transfer capture*  $\theta \in [0, 1]$  is the share of a book the non-incumbent retains after taking over the zip — attrition and persistency account for  $1 - \theta$ . *Headroom credit*  $\lambda \in [0, 1]$  is how much a dollar of unwritten opportunity counts against a dollar of booked production.

Headroom is taken *net*: whoever is assigned  $z$  is credited with opportunity net of business already on the books there, not gross of it. Gross double-counts and is incoherent whenever  $\lambda > \theta$  — it would make an inherited book of business a *liability*, since the same dollar of production is worth more credited as headroom than credited as a captured book. Under the net convention, with

$$c_1 = 1 - \lambda, \quad c_2 = \theta(1 - \lambda), \quad (1)$$

the value of zip  $z$  to each wholesaler, were it assigned the whole territory, is

$$u_a(z) = c_1 A_z + c_2 B_z + \lambda M_z, \quad u_b(z) = c_2 A_z + c_1 B_z + \lambda M_z. \quad (2)$$

$u_a$  credits wholesaler  $a$  with all of their own book, a  $\theta$ -share of  $b$ 's, and a  $\lambda$ -share of what neither has written;  $u_b$  is the mirror image. Net headroom is feasible only if it is non-negative pointwise,

$$M_z \geq \max(A_z + \theta B_z, B_z + \theta A_z) \quad \text{for every } z, \quad (3)$$

which we take as maintained throughout. An *allocation* is a partition of  $Z = \{1, \dots, n\}$  into a set  $S$  given to wholesaler  $a$  and its complement  $Z \setminus S$  given to  $b$ : every zip is assigned, to exactly one side, with no third disposition.

**Related work.** The problem sits at the meet of two literatures that rarely cite each other. Sales-territory alignment and districting build the optimisation machinery: the assignment formulation of Hess and Samuels (1971), its sales-management line (Zoltners and Sinha, 1983; Zoltners and Lorimer, 2000; Zoltners and Sinha, 2005), the modern exact treatments of territory design (Salazar–Aguilar et al., 2011; Ríos-Mercado and Pérez, 2012), and the contiguity encodings compared in Section 2 — but it optimises managerial objectives, not fairness between the two parties. Fair division on graphs supplies the fairness theory — allocations restricted to connected pieces (Bouveret et al., 2017), existence of connected EF1 divisions (Bilò et al., 2022; Suksompong, 2019), and the price of connectivity (Bei et al., 2022) — but rarely an exact solver at data scale. This paper takes the criterion from the second literature and the machinery from the first.

## 2 The bargaining problem

This section poses territory division as a two-person bargaining problem in the classical sense (Nash, 1950), then works down from that object to the program actually solved. Seven layers.

### 2.1 Layer 1 — players, outcomes, and the feasible set

The players are the two wholesalers,  $a$  and  $b$ . An outcome is an allocation  $S \subseteq Z$  as defined above, giving each player the payoff

$$U_a(S) = \sum_{z \in S} u_a(z), \quad U_b(S) = \sum_{z \notin S} u_b(z), \quad (4)$$

using the utilities (2). The *feasible set*

$$F = \{(U_a(S), U_b(S)) : S \subseteq Z\} \subset \mathbb{R}^2 \quad (5)$$

is the set of attainable payoff pairs: a cloud of at most  $2^n$  points, entirely determined by the data  $(A_z, B_z, M_z)$  and the parameters  $\theta, \lambda$ .

### 2.2 Layer 2 — the disagreement point

A bargaining problem is a pair  $(F, d)$ , where  $d \in \mathbb{R}^2$  is the *disagreement point*: the payoff each player secures unilaterally, without the other’s cooperation, if no allocation is agreed. Getting  $d$  right is not a normalisation — it is a claim about what each wholesaler can actually walk away with, and it is the single most consequential modelling choice in this paper.

An earlier draft set  $d = (S_a, S_b)$ , each wholesaler’s pre-merger production: if the division fails, each keeps the book they walked in with. This is the natural first guess, and it is wrong. A disagreement point must be an outcome a player can execute *unilaterally* if bargaining breaks down. Once the merger closes, pre-merger production is not such an outcome: the separate sales forces, appointments, and client relationships that generated  $S_a$  and  $S_b$  are dissolved into the merged entity, and neither wholesaler can unilaterally resurrect them regardless of how the territory dispute resolves.  $S_a$  is a historical number, not a threat point. What each wholesaler can secure by failing to agree is nothing: absent an assignment, neither is appointed or staffed to write business anywhere in the disputed territory. The disagreement point is therefore the origin,

$$d = (0, 0). \quad (6)$$

**This is not a matter of taste.**  $d = (S_a, S_b)$  with  $S_a \neq S_b$  is not a neutral rescaling: bargaining from an asymmetric threat point is observationally equivalent to an *asymmetric* Nash solution at  $d = 0$  with an implicit, undisclosed weight (here  $\omega \approx 0.54$  on the worked instance) — a tilt toward whichever side had the bigger pre-merger book, smuggled in through the disagreement point rather than argued for on its merits. It also forfeits envy-freeness up to one good (EF1): the Nash bargaining solution is EF1 at  $d = 0$  (Caragiannis et al., 2019) and the guarantee does not survive moving  $d$  away from the origin. Both defects disappear at  $d = (0, 0)$ .

Writing  $g_a(S) = U_a(S) - 0$  and  $g_b(S) = U_b(S) - 0$  for the gains relative to  $d$ , gains are simply bundle utilities:

$$g_a(S) = \sum_{z \in S} u_a(z), \quad g_b(S) = \sum_{z \notin S} u_b(z), \quad G_a^{\max} = \sum_z u_a(z), \quad G_b^{\max} = \sum_z u_b(z), \quad (7)$$

with  $G_a^{\max}, G_b^{\max}$  the attainable maxima from receiving the whole territory. Because  $\theta, \lambda \in [0, 1]$  and (3) holds,  $u_a(z), u_b(z) \geq 0$  pointwise, so  $g_a(S) > 0$  for every non-empty  $S$  and  $g_b(S) > 0$  for every  $S \neq Z$ : unlike the pre-merger baseline, where headroom below a threshold  $\lambda^*$  left the bargaining set empty, feasibility of both gains is automatic here for any non-trivial split. That threshold was an artefact of the discarded convention and does not arise at  $d = 0$ .

### 2.3 Layer 3 — Nash’s axioms and the solution

Nash (1950) shows that if  $F$  is convex and compact and contains a point strictly dominating  $d$ , exactly one point of  $F$  satisfies four axioms:

**Pareto optimality.** No other feasible point makes both players weakly better off and one strictly better off.

**Symmetry.** If  $F$  is symmetric in the two players and  $d_a = d_b$ , the solution gives them equal payoff.

**Invariance to affine transformations.** The solution commutes with independently rescaling and shifting either player’s utility — utility is cardinal only up to a positive affine transformation, and the solution must not depend on the choice of units.

**Independence of irrelevant alternatives.** Shrinking  $F$  to a subset that still contains the solution does not change it.

That point is the maximiser of the *Nash product*, the product of gains over the disagreement point:

$$\max_{(U_a, U_b) \in F} (U_a - d_a)(U_b - d_b) \quad \text{subject to } U_a > d_a, U_b > d_b. \quad (8)$$

No other point of  $F$  satisfies all four axioms simultaneously; this is what makes (8) a characterisation rather than one criterion among several equally defensible ones.

**$F$  is not convex here.** Nash’s theorem is stated for convex, compact  $F$ ; with indivisible postal codes  $F$  is a finite point cloud instead, so strictly the axioms select a *lottery* over two adjacent allocations, and (8) maximised over the point cloud is the discrete analogue rather than the axiomatic solution itself. The gap is quantified in the remark on convexity in Section 3: 0.0045% on the worked instance. Known, small, ignored in practice; worth re-checking on real data.

## 2.4 Layer 4 — specializing to the discrete allocation problem

At  $d = (0, 0)$ , applying (8) to the feasible set generated by allocations gives the criterion actually solved throughout this paper:

$$\max_{S \subseteq Z} g_a(S) g_b(S), \quad g_a(S) > 0, \quad g_b(S) > 0, \quad (9)$$

with  $g_a, g_b$  as in (7). In the fair-division literature this same criterion is *maximum Nash welfare*: the two literatures meet at  $d = 0$ , where the bargaining solution and the welfare maximiser coincide, and the fair-division results — Pareto efficiency and envy-freeness up to one good over indivisible items (Caragiannis et al., 2019), and the study of allocations constrained to connected pieces of a graph (Bouveret et al., 2017) — apply to the bargaining problem verbatim. The remainder of this section turns (9) into a program that can be handed to a solver: first restricting  $F$  by operational constraints (contiguity, state borders), then writing the result as a mixed-integer program, then solving it exactly.

## 2.5 Layer 5 — constraints shrink $F$

Operational requirements restrict which  $S$  are admissible, and therefore which points of  $F$  are attainable. Contiguity — each wholesaler receiving one connected territory on the postal-code adjacency graph  $G = (Z, E)$  — is the binding one:

$$F_{\text{cont}} = \{(g_a(S), g_b(S)) : G[S] \text{ and } G[Z \setminus S] \text{ both connected}\} \subseteq F. \quad (10)$$

The same bargaining problem is then solved on the smaller set. *That is the precise game-theoretic content of the districting constraints: they do not change the criterion, they shrink the feasible set.* State-border restrictions act identically, by deleting cross-state edges from  $G$  before connectivity is assessed.

## 2.6 Layer 6 — the program

Maximising (8) over  $F_{\text{cont}}$  at  $d = (0, 0)$ , with  $x_z = \mathbf{1}[z \in S]$ :

$$\max_{x \in \{0,1\}^n} \log g_a + \log g_b \quad (11)$$

$$\text{s.t.} \quad g_a = \sum_z x_z u_a(z), \quad g_b = \sum_z (1 - x_z) u_b(z), \quad (12)$$

$$g_a > 0, \quad g_b > 0, \quad (13)$$

$$G[S] \text{ and } G[Z \setminus S] \text{ connected.} \quad (14)$$

**This is a convex MINLP.** The gains (12) are *linear* in  $x$ ; the objective (11) is *concave* in those linear expressions; connectivity (14) is representable by linear constraints. Maximising a concave function over a mixed-integer linear feasible set is the well-behaved MINLP class, for which outer approximation is a finite global method. The product form  $\max g_a g_b$  is *not* concave — it is bilinear and indefinite — so the logarithm is not cosmetic: it is what places the problem in the tractable class.

## 2.7 Two representations of connectivity

Constraint (14) is not a linear inequality as written. Two standard encodings.

(a) **Compact, single-commodity flow.** (Shirabe, 2005, 2009) Fix roots  $r_a, r_b$ . Introduce  $f_{uv}^a \geq 0$  on each arc. Every zip assigned to  $a$  absorbs one unit of flow sent from  $r_a$  through arcs internal to  $a$ 's territory:

$$\sum_{(u,z) \in E} f_{uz}^a - \sum_{(z,v) \in E} f_{zv}^a = x_z, \quad z \neq r_a, \quad (15)$$

$$f_{uv}^a \leq (n-1)x_u, \quad f_{uv}^a \leq (n-1)x_v, \quad (u,v) \in E, \quad (16)$$

with the mirror system in  $(1-x)$  for  $b$ . Polynomial size,  $O(|E|)$  continuous variables per district, one solve.

(b) **Exponential, separator cuts.** For every  $S' \subseteq Z$  inducing a component not containing the root,

$$|S'| \sum_{w \in N(S')} x_w \geq \sum_{v \in S'} x_v. \quad (17)$$

There are exponentially many such inequalities, so they are generated *lazily*: solve, inspect the incumbent for disconnected components, add the violated cuts, re-solve. In practice a handful of rounds suffice. This is the encoding used here; it keeps the model small at the cost of repeated solves.

The two encodings, a multicommodity-flow strengthening, and the separator-cut family are compared head-to-head in the districting literature: Duque et al. (2011) for the one-shot formulations, Validi et al. (2021); Validi and Buchanan (2022) for lazily separated cuts inside branch-and-cut, which certify instances of  $\sim 1,500$  units and are the state of the art. A linear-size planar-graph formulation exists (Zhang et al., 2024) but ties the model to a planar embedding; ZCTA adjacency is near-planar, not planar, so it is not used here. How the lazy scheme is best organised — one restarted MILP per round versus one branch-and-cut tree fed cuts through callbacks (Quesada and Grossmann, 1992; Bonami et al., 2008) — turns out to matter more than the encoding; Section 6.2 returns to this.

## 2.8 Layer 7 — solving it

Outer approximation linearises the concave objective from above. Introduce  $z_a \leq \log g_a$ ,  $z_b \leq \log g_b$  enforced by supporting hyperplanes (18), and maximise  $z_a + z_b$ . Each relaxation is a pure MILP; its optimum is an upper bound on (11); a tangent added at the incumbent tightens it. Interleaving the tangent cuts with the connectivity cuts (17) converges both simultaneously, and termination when the incumbent's true log-product meets the bound is an optimality *certificate*. Compactness may be priced by subtracting  $\rho \sum_e y_e$  with  $y_e \geq |x_u - x_v|$ , at the cost of changing the objective.

| layer | object                                  | form                                               |
|-------|-----------------------------------------|----------------------------------------------------|
| 1     | players, outcomes, feasible set $F$     | $2^n$ points                                       |
| 2     | disagreement point                      | $d = (0, 0)$                                       |
| 3     | Nash’s axioms $\Rightarrow$ solution    | $\max(U_a - d_a)(U_b - d_b)$                       |
| 4     | discrete specialisation                 | $\max g_a(S)g_b(S)$                                |
| 5     | constraints (contiguity, state borders) | shrink $F$                                         |
| 6     | program                                 | (11)–(14), convex MINLP                            |
| 7     | method                                  | outer approximation + lazy cuts, finite, certified |

### 3 The Nash solution

Section 2 reduced fair division to the criterion (9),  $\max_{S \subseteq Z} g_a(S)g_b(S)$  subject to  $g_a, g_b > 0$ . This section gives the exact algorithm. A fast, solver-free heuristic based on ordering zips by exchange rate — useful as a preview or sanity check, but not exact for Nash — is developed in Appendix B.

Taking logarithms,  $\log(g_ag_b) = \log g_a + \log g_b$ , does not linearise the objective:  $g_a$  is linear in  $x$ , so  $\log g_a$  is *concave* in a linear expression. The problem is therefore a convex MINLP — maximising a concave function over a MILP-representable set — for which outer approximation is a finite *global* method. Introduce  $z_a \leq \log g_a$  and add supporting hyperplanes

$$z_a \leq \log(\hat{g}) + \frac{g_a - \hat{g}}{\hat{g}}, \quad (18)$$

one linear cut per tangent point, no additional binaries, and maximise  $z_a + z_b$ . The tangent set is a relaxation, so the MILP optimum is a valid *upper bound*; adding a tangent at the incumbent and re-solving tightens it, and when the incumbent’s true log-product meets the bound the incumbent is globally optimal. This is a certificate, not a stopping heuristic.

Validated against exhaustive enumeration, 120 random instances per size: exact match in 100% of cases at every  $n$  tested, with bound gap identically zero. Cost is 6–7 iterations regardless of size — 0.10s at  $n = 50$ , 0.84s at  $n = 400$ . **Use the exact method by default**; the prefix table (Appendix B) is a fast preview and a sanity check.

*Remark* (Convexity, and what “Nash” means over indivisible zips). Nash’s axiomatic characterisation — Pareto efficiency, symmetry, scale invariance, independence of irrelevant alternatives — assumes the bargaining set is *convex*. With indivisible postal codes the feasible set  $F = \{(g_a(S), g_b(S))\}$  is a finite point cloud, so strictly the axioms deliver a *lottery* over two adjacent allocations, and  $\max g_ag_b$  over  $F$  is the natural discrete analogue rather than the axiomatic solution itself. The distinction is quantitatively small here: the Nash product on the convex hull, where randomisation is permitted, is 53.30998 against 53.30759 for the best deterministic allocation — a difference of 0.0045%. The frontier is dense enough that the deterministic vertex lies essentially on the hull. It is worth re-checking on real data, and not worth randomising territory assignments over.

*Remark* (No fixed point is needed). In a continuum the Nash threshold satisfies  $t^* = g_a/g_b$ , a fixed point whose naive iteration diverges — on this instance it runs  $1.000 \rightarrow 40.22 \rightarrow -0.244 \rightarrow -6.447$  into a degenerate corner. Neither the prefix table nor the outer approximation requires it.

## 4 Why Nash

**Exactly solvable.** Outer approximation (Section 3) reaches a certified global optimum in 6–7 iterations regardless of  $n$ . The equalisation criteria require search for their exact optima and carry no such certificate.

**Pareto efficient by construction.** Maximising a product cannot be improved for both parties simultaneously. The min-|difference| criteria carry no such guarantee and do fail in practice: on the worked instance the unconstrained equalising program achieves near-perfect equality at gains of (7.323, 7.070) — *both* strictly below the Nash allocation’s (7.371, 7.232). Equalisation purchased fairness by making both wholesalers worse off (Appendix A).

**Envy-freeness up to one zip.** Neither wholesaler prefers the other’s territory once a single zip is removed from it — the strongest fairness guarantee available for indivisible goods, and legible in a room. The guarantee is a theorem for the *unconstrained* maximiser (Caragiannis et al., 2019); under contiguity the picture is more delicate. A *connected* EF1 allocation for two agents exists on any graph whose block-tree is a path (Bilò et al., 2022, Thm. 3.10) — every biconnected census component qualifies — and is computable in  $O(m)$  by a discrete cut-and-choose. Whether the contiguous *Nash maximiser* is itself EF1 for two agents is open: the known counterexamples to EF1 on graphs use three or more agents (Igarashi and Peters, 2019). We therefore report EF1 empirically alongside every contiguous solve; it held on every certified incumbent in the 2026 benchmark runs.

**The continuum limit agrees.** Warren (2025) studies Nash bargaining over a continuum of goods and finds that at a zero disagreement point the solution is an optimal-transport map, whose allocation boundaries are Laguerre cells — contiguous by construction. That corroborates  $d = (0, 0)$  from a direction independent of the axiomatic argument of Section 2: the correspondence is stated at the zero baseline, and the contiguity that must be imposed here as a hard constraint (Section 6) comes for free there. One caution about importing the geometry: the continuum contiguity is produced by *distance-structured* utilities — the power-diagram cells inherit their shape from a spatial cost — and has no analogue for data-driven per-zip utilities, where the transport solution is exactly the free ratio-threshold allocation and no more connected than it. Contiguity would become structural rather than imposed only under a modelling change, an explicit travel cost  $-\kappa d(z, p_i)$  in the utilities, which we treat as an open modelling experiment, not a solver device. The problem solved in this note is a finite sampling of that continuum, so the agreement is a consistency check rather than a corollary.

**If seniority should count.** Asymmetric Nash,  $\max g_a^\omega g_b^{1-\omega}$ , retains the prefix property with threshold  $\frac{1-\omega}{\omega} \cdot \frac{g_a}{g_b}$ . An explicit weight is preferable to a criterion that encodes a tilt nobody selected.

## 5 A fifty-zip instance

Fifty postal codes on the unit square drawn from a two-component Gaussian mixture, with metros at (0.28, 0.62) and (0.72, 0.36);  $a$  is anchored on the first,  $b$  on the second, both present in each. Totals  $S_a = 3.0$ ,  $S_b = 1.8$ ,  $M = 40$  (12% combined penetration);  $\theta = 0.4$ ,  $\lambda = 0.3$ .

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|                                     |                  |
|-------------------------------------|------------------|
| $\text{corr}(A_z, B_z)$ across zips | +0.685           |
| utility ratio range                 | [0.9531, 1.1373] |
| minimum net headroom                | 0.2335           |

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The positive correlation is the harder configuration: both wholesalers are strong in the same places. The ratio spans a narrow band mostly above 1, so on raw utility  $a$  is favoured nearly everywhere and fairness is the only force conceding territory to  $b$ .

50 zips on the unit square — Voronoi cells coloured by field; marker size  $\propto$  magnitude; stars mark the two metro centres

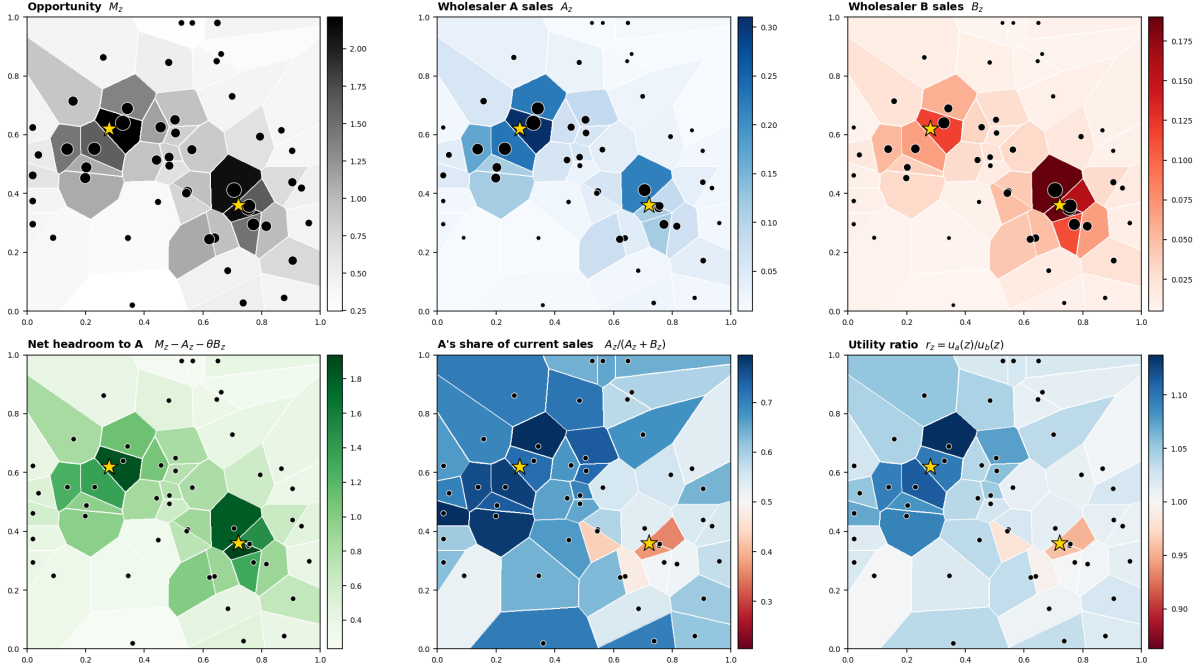


Figure 1: The instance before optimisation. Voronoi cells coloured by field, marker size proportional to magnitude, stars at metro centres.

## 5.1 Solution

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|                                     | $k$ | $g_a$  | $g_b$  | $g_a g_b$       |
|-------------------------------------|-----|--------|--------|-----------------|
| Nash, exact (outer approximation)   | 22  | 7.3715 | 7.2318 | <b>53.30907</b> |
| Nash, prefix heuristic (Appendix B) | 22  | 7.4423 | 7.1628 | 53.30759        |

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The two select *different* allocations of the same size; the heuristic’s shortfall is  $2.8 \times 10^{-5}$ , which is negligible here but not zero.

$a$  receives 22 zips, 48.9% of opportunity. Kalai–Smorodinsky (Kalai and Smorodinsky, 1975) and equal opportunity coincide exactly with Nash here; equal gain and the egalitarian criterion differ by one zip (Appendix A).



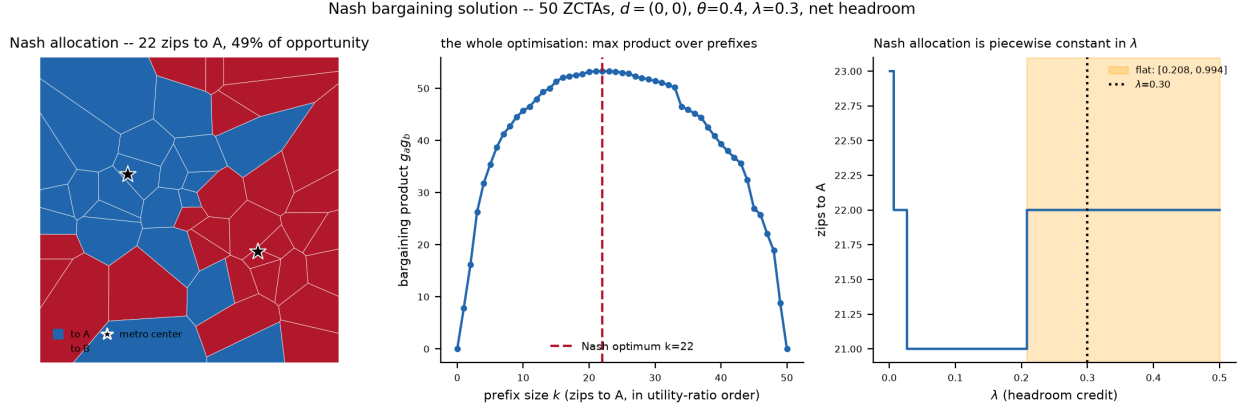


Figure 2: Left: the Nash allocation. Centre: the entire optimisation — the bargaining product over the  $n + 1$  prefixes, maximised at  $k = 22$ . Right: the allocation is piecewise constant in  $\lambda$ , with  $\lambda = 0.30$  inside a flat interval running from 0.208 to at least 0.994.

## 6 Contiguity

Nothing above forces each territory to be connected, and the Nash allocation is not:  $a$  is already a single contiguous block, but  $b$  holds a four-zip island. Imposing connectivity turns the problem into a *districting* integer program — the formulation used for electoral redistricting and sales-territory alignment — with one binary per zip.

### 6.1 Formulation: the log makes it concave, not linear

Taking logarithms separates the Nash product,  $\log(g_a g_b) = \log g_a + \log g_b$ , but does not linearise it:  $g_a$  is linear in  $x$ , so  $\log g_a$  is *concave* in a linear expression. That is the useful case. A concave function is the lower envelope of its tangents, so introduce  $z_a \leq \log g_a$  and add supporting cuts

$$z_a \leq \log(\hat{g}) + \frac{g_a - \hat{g}}{\hat{g}}, \quad (19)$$

one linear constraint per tangent point  $\hat{g}$ , with no additional binaries. The program is then

$$\text{decision } x_z \in \{0, 1\}, \text{ 1 if } z \rightarrow a, \quad (20)$$

$$\text{objective } \max z_a + z_b - \rho \sum_{e \in E} y_e, \quad y_e \geq |x_u - x_v|, \quad (21)$$

with  $z_a, z_b$  bounded by tangent cuts (19). The tangent set is an *outer* approximation — an upper bound — which tightens as tangents are added at the incumbent solution. This is Kelley’s cutting-plane method, and it interleaves naturally with the contiguity cuts of Section 6.2: both converge together, typically in about five iterations.

The  $y_e$  count *boundary edges*, the perimeter of the partition, and  $\rho$  prices compactness. Note that  $\rho > 0$  *changes the objective*: the solve is then exact for Nash-minus-a-perimeter-penalty, a legitimate modelling choice but not exact Nash. Set  $\rho = 0$  for the latter.

**Three traps.** First, in the restart-per-round loop the compactness term is not optional: fairness alone is degenerate, many allocations achieve near-identical objective values, and contiguity cuts never bite — an implementation without it thrashed for seventeen rounds without converging.

This is textbook instability of Kelley’s method (Lemaréchal et al., 1995); the principled remedies are in–out stabilisation of the cut loop (Ben-Ameur and Neto, 2007; Fischetti and Salvagnin, 2010) or moving to a single search tree (Section 6.2), after which  $\rho = 0$  — exact Nash, the model of record — solves cleanly and  $\rho > 0$  is no longer needed as an engineering crutch. Second, a weighted-sum scalarisation  $wg_a + (1 - w)g_b$  collapses to corner solutions when the frontier is close to linear, as it is here, and an  $\varepsilon$ -constraint sweep needs one MILP solve per frontier point. Third, the tangent cuts and the connectivity cuts obey *different* finite-convergence theories — outer approximation’s (Duran and Grossmann, 1986; Fletcher and Leyffer, 1994) versus combinatorial Benders’ (Hooker and Ottosson, 2003; Codato and Fischetti, 2006) — and in a restarted loop their iteration bounds multiply rather than add. An alternative that removes the tangent loop entirely is an  $\varepsilon$ -certified piecewise-linear encoding of the logarithm (Caragiannis et al., 2019, §6), with the standard MILP encodings of Vielma and Nemhauser (2011); it trades the certificate’s exactness for a fixed  $\varepsilon$ .

## 6.2 Contiguity by separator cuts

Fix a root zip per side. If  $a$ ’s territory splits, then for each component  $S$  not containing  $a$ ’s root add

$$|S| \sum_{w \in N(S)} x_w \geq \sum_{v \in S} x_v, \quad (22)$$

“if any of  $S$  is with  $a$ , some neighbour just outside must be too”; symmetric in  $(1 - x)$  for  $b$ . Re-solve and repeat. The alternative is a single-commodity *flow* formulation — ship one unit from each assigned zip to its root along in-district edges (Shirabe, 2005) — exact in one solve but requiring  $O(|E|)$  extra continuous variables per district. Aggregating one cut per island rather than one per zip matters for convergence.

**Rooted cuts are invalid on a disconnected instance.** When the contested region itself has several graph components — routine on real geography — a rooted cut can force growth toward a root that is unreachable, excluding allocations that are feasible under the right notion, *component-wise* contiguity. The master problem is then no longer a relaxation and its “bound” can sit below feasible values already seen. The production formulation therefore uses *root-free* pairwise cuts: for  $u, v$  on the same side and  $C$  a minimal  $u, v$ -separator,  $\sum_{w \in C} x_w \geq x_u + x_v - 1$ , the vertex-separator family of Validi et al. (2021); Validi and Buchanan (2022) with the root dependence removed. Any  $u, v$  and any separator give a valid cut, which also makes separation at fractional points safe.

How the cuts are delivered matters as much as their form. The restart-per-round (multi-tree) loop described above re-proves the same tree from scratch each round; the state of the art places both cut families — log tangents and separator cuts — as lazy constraints inside *one* branch-and-cut tree (Quesada and Grossmann, 1992; Bonami et al., 2008; Validi et al., 2021), which is the architecture the benchmark of Section 6.4 converged on. Its engine must expose a cut-injection callback; of the open-source stack, SCIP does and HiGHS does not.

## 6.3 Result

The exact unconstrained Nash allocation leaves  $a$  in one piece and  $b$  in two, a 24-zip block plus a stray 4-zip island. Enforcing connectivity, and sweeping the compactness weight on a graph of 50 zips and 132 adjacency edges:

| $\rho$             | $g_a g_b$ | perimeter | % of edges | iterations | time | pieces $a/b$ |
|--------------------|-----------|-----------|------------|------------|------|--------------|
| — (unconstrained)  | 53.3091   | —         | —          | —          | —    | 1/2          |
| $2 \times 10^{-3}$ | 53.2046   | 15        | 11.4%      | 4          | 0.2s | 1/1          |
| $2 \times 10^{-4}$ | 53.2769   | 21        | 15.9%      | 6          | 0.3s | 1/1          |
| $1 \times 10^{-5}$ | 53.2918   | 27        | 20.5%      | 9          | 0.1s | 1/1          |

The price of contiguity under Nash is between 0.03% and 0.20% of the bargaining product, depending on how clean a boundary is wanted. Contiguity is close to free when solved exactly. A greedy island-repair heuristic, by contrast, degraded the objective by more than three orders of magnitude and should be used only to *detect* fragmentation, never to fix it.

$\rho$  competes directly with the log objective: at  $\rho = 10^{-5}$  the boundary cuts a fifth of all adjacency edges for a 0.16% gain in product over the  $\rho = 2 \times 10^{-3}$  solution, so if a positive  $\rho$  is used it should be swept and reported with the perimeter. The model of record, however, is  $\rho = 0$  — exact Nash under hard contiguity, with ties broken by a lexicographic perimeter post-pass rather than a penalty; positive  $\rho$  was a stabilisation crutch for the restarted loop that the single-tree architecture (Section 6.4) does not need.

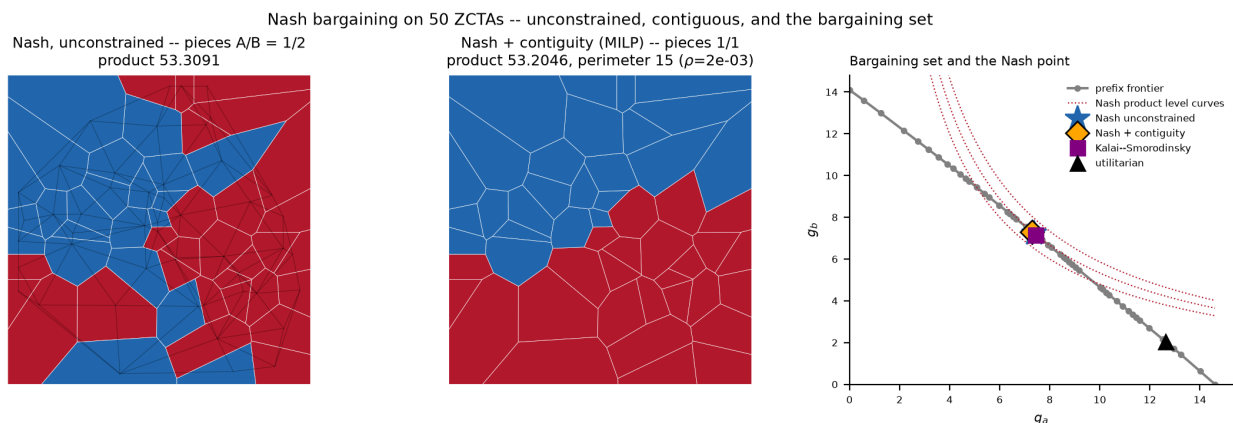


Figure 3: Left: unconstrained Nash with the Delaunay adjacency overlaid;  $a$  is already one block,  $b$  holds a four-zip island. Centre: the contiguous MILP solution. Right: the bargaining set — grey points are the prefix frontier, dotted curves are level sets of  $g_a g_b$ , and the Nash point is where the highest attainable level curve touches the frontier. The utilitarian corner leaves  $b$  with a gain of 2.04.

## 6.4 Failure mechanisms at scale

The fifty-zip instance solves in a fraction of a second; real bilateral pairs run to 400–800 ZCTAs. A synthetic battery — varying territory alignment, book correlation, state borders, headroom tightness, and the tail weight of the value distributions — shows that the restart-per-round loop fails to converge for three *independent* reasons, so a fix aimed at one is not guaranteed to touch the others:

1. **Pre-existing disconnection.** The contested region itself arrives in several graph components,

and rooted cuts are then unsound (the warning in Section 6.2). This is a formulation defect, fully repaired by root-free component-wise cuts — not a solver limitation.

2. **Pure scale.** Beyond roughly 125 zips the restarted loop exhausts time regardless of topology; the single-tree architecture is the repair.
3. **Value concentration.** Replacing lognormal by heavy-tailed (double Pareto-lognormal) sales noise — same geometry, same adjacency — *moves which pairs fail*: a previously trivial 31-zip pair stalls while a previously failing disconnected pair converges, and the cost of contiguity inflates two- to five-fold. The analogy with knapsack hardness under correlated, concentrated weights (Pisinger, 2005) is apt: difficulty tracks value structure, not size.

Real data adds a fourth regime the battery cannot: *sparse active zips* — a minority of zips carrying nearly all value, glued by many near-zero ones. That regime is addressed by safe reduction and variable-fixing techniques from the Steiner-tree and MILP-presolve literature (Rehfeldt et al., 2019; Buchanan et al., 2018; Fischetti et al., 2017), applied before the tree is grown.

In the 2026 benchmark of these repairs, the single search tree (SCIP) with root-free separator cuts and the native concave logarithm certifies every battery pair up to  $\sim 125$  zips in under a second; from 125 to 464 zips it holds certified gaps between  $10^{-7}$  and  $4 \times 10^{-3}$  nats with the time budget genuinely spent. The residuals have two distinct characters. Up to  $\sim 160$  zips the residual is the LP feasibility *tolerance floor* — the solver reaches its own optimum on a loosened rung and nothing tighter is numerically available; an exact post-hoc bound, not more time, is the cure. Above that the residual is *dual slack over near-ties*: an iterated local search finds, in seconds, an allocation that independent restarts agree on to nine digits, and the tree warm-started *at that allocation* still cannot prove it within the budget — the bound, not the incumbent, is the expensive half, and its difficulty tracks value concentration (mechanism 3), not  $n$ .

Certification is therefore two-tier. Small and mid-size pairs carry exact certificates (gap  $\leq 10^{-8}$  nats). At production size a pair is accepted at gap  $\leq \varepsilon = 5 \times 10^{-3}$  nats together with cross-method agreement on the allocation itself. The  $\varepsilon$  is not a concession chosen for convenience but is calibrated to the data: re-drawing the books under the measurement-noise model of Section 7 moves the *optimal* log-product by  $8.4 \times 10^{-3}$  nats ( $1\sigma$ ) and reassigns one zip in eight, so  $\varepsilon$  bounds the unproven improvement strictly below what the data can distinguish.

**The price of connectivity.** Across the battery the cost of contiguity ranges from 0.03% to 5% of the bargaining product, typically  $\sim 0.4\%$  at scale — against which the known theory is: for maximin-share fairness the worst-case price of connectivity is exactly  $4/3$  on biconnected graphs and  $1/k$  at a  $k$ -way cut vertex (Bei et al., 2022); for Nash welfare no bound is published, and we state it as open. Connected fair division is NP-hard already for two agents with *identical* valuations (Deligkas et al., 2021), which is what justifies reaching for an exact integer program rather than a rule.

## 7 Contestability

Where are the hard decisions? Two components, multiplied.

**Stake.** Misassigning zip  $z$  moves the bargaining product by, to first order,

$$\text{stake}_z = \frac{u_a(z)}{g_a} + \frac{u_b(z)}{g_b}, \quad (23)$$

the relative swing in  $\log g_a g_b$ . On the instance it ranged over  $[0.0115, 0.1216]$ .

**Doubt.** The flip rate  $2 \min(p, 1-p)$  under uncertainty. Sweeping  $\theta \in [0.20, 0.60]$  and  $\lambda \in [0.10, 0.50]$  alone moves 3 of 50 zips; perturbing the *data* by 10% on sales and 6% on opportunity moves 32 of 50.

Parameter uncertainty is nearly irrelevant here; data uncertainty dominates. Getting  $A_z$  and  $B_z$  right matters more than settling  $\lambda$ .

| rank | zip | stake  | doubt | quadrant   |
|------|-----|--------|-------|------------|
| 1    | 37  | 0.0541 | 0.613 | adjudicate |
| 2    | 32  | 0.1159 | 0.243 | adjudicate |
| 3    | 22  | 0.0471 | 0.540 | adjudicate |
| 4    | 19  | 0.0352 | 0.540 | adjudicate |
| 5    | 28  | 0.0302 | 0.567 | coin-flip  |

Splitting stake and doubt at their medians, nine zips fall in the adjudicate quadrant; the top five carry 52% of total contestability. Zip 37 carries the largest doubt at moderate stake and zip 32 the largest stake at moderate doubt — the two biggest open questions are different zips for different reasons, not one dominant dispute.

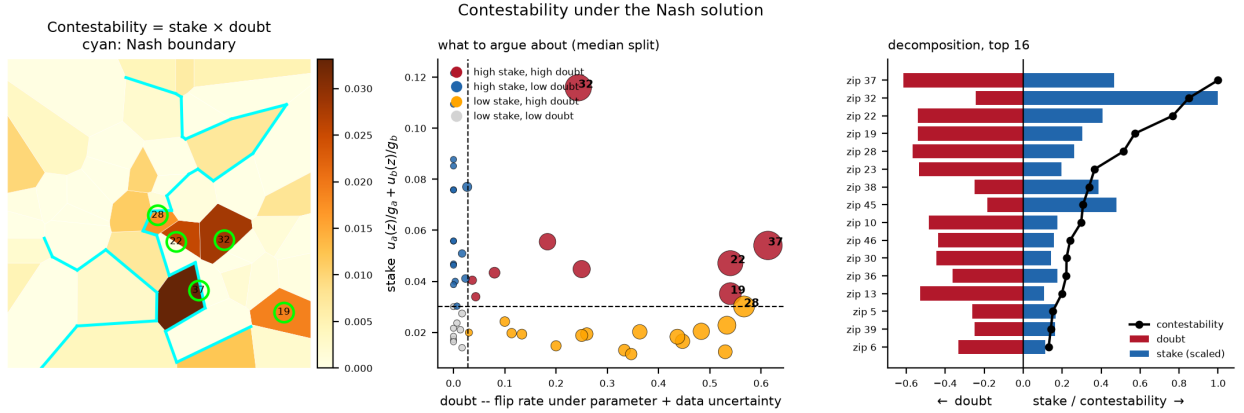


Figure 4: Contestability as stake  $\times$  doubt. Left: mapped, Nash boundary in cyan, top five ringed. Centre: the  $2 \times 2$ . Right: decomposition of the top sixteen.

## 8 Exact parameter breakpoints

Write  $P_z = A_z + \theta B_z$ ,  $Q_z = B_z + \theta A_z$ , so  $u_a(z) = (1 - \lambda)P_z + \lambda M_z$ . Zips  $i, j$  exchange places in the ratio order exactly when

$$(1 - \lambda)K_{ij} + \lambda L_{ij} = 0 \implies \lambda_{ij}^* = \frac{K_{ij}}{K_{ij} - L_{ij}}, \quad \begin{aligned} K_{ij} &= P_i Q_j - P_j Q_i, \\ L_{ij} &= P_i M_j - P_j M_i + Q_j M_i - Q_i M_j. \end{aligned} \quad (24)$$

One explicit expression per pair; in  $\theta$  the condition is quadratic, hence also closed form. This ordering is unaffected by the disagreement point —  $r_z = u_a(z)/u_b(z)$  does not involve  $d$  — but which prefix the Nash criterion selects does. On the instance at  $d = (0, 0)$  the Nash allocation takes only **two distinct values** across  $\lambda \in [0.02, 0.90]$ : 21 zips on a narrow band, 22 elsewhere.

$\lambda = 0.30$  sits inside a flat interval from 0.208 to at least 0.994: anyone who believes a dollar of untapped opportunity is worth between roughly 21 cents and essentially all of a dollar of booked production obtains the same map. All of the switching happens in a narrow band,  $\lambda \in (0.026, 0.208)$ .

This is a stronger instrument than a local derivative: rather than reporting  $dk/d\lambda$ , one states the complete set of possible answers and exactly which belief produces each.

## 9 Opportunity balance

Before the merger both wholesalers covered the whole market, each with access to  $M = 40$  at penetrations of 7.5% and 4.5%. The honest framing is that both end with roughly half of what they had. Under Nash,  $M_a = 19.57$  (48.9%) against  $M_b = 20.43$ , an imbalance of 2.2%.

**$\lambda$  is the opportunity-balance dial.** The headroom term places most of the weight on opportunity at moderate  $\lambda$ , so the criterion balances market size without being asked to. Here the effect is a single step rather than a smooth decline: imbalance sits at 4.4% for  $\lambda \lesssim 0.21$  and drops to 1.1% once  $\lambda \gtrsim 0.21$ , tracking the same breakpoint identified in Section 8. “How much should untapped opportunity count?” and “should territories carry equal market size?” are the same question.

Imposing equal opportunity as a hard condition is possible but is a strong position: with books of 3.0 against 1.8 it fully discounts the larger book. That is a compensation-philosophy decision rather than a modelling one.

## 10 Implementation and scope

`territory.py` operates on a `networkx` graph of ZCTAs with Rook adjacency and attributes `rep_a`, `rep_b`, `A`, `B`, `M`, `state`: it validates headroom, builds the bipartite overlap graph of wholesaler pairs and reports its component structure, solves each pair exactly by outer approximation (the prefix table remains as a preview), and writes results back for mapping. `districting.py` adds the contiguous MILP. Setting `respect_state=True` deletes cross-state edges before enforcing connectivity; for life and annuity distribution this may bind harder than contiguity, since product availability and producer appointments are state-scoped.

**Not covered.** Capacity and travel constraints. Three or more wholesalers, for which no threshold rule exists and the natural extension is leximin over each dense component of the overlap graph; Warren (2025) points to a second route, since semi-discrete transport to  $N$  sites yields  $N$  Laguerre cells and would deliver the multi-party division and its geography together rather than in two stages. Estimation of  $\theta$  from historical reassignments, on which everything ultimately rests. At national scale the production stack is open-source single-tree branch-and-cut: PySCIPOpt as the primary engine (SCIP exposes the constraint-handler callback that lazy separator and tangent cuts require; HiGHS has no cut-injection callback, which is why the restart-per-round loop was structural), with python-mip/CBC as an independent cross-check. The benchmark harness behind Section 6.4 — instance tiers, gap metrics, and acceptance criteria, including the two-tier certificate — is specified in `research/contiguity/TEST_PLAN.md` of the accompanying repository.

### Three checks on real data.

1.  $\text{corr}(A_z, B_z)$  across zips — whether the books overlap or separate, and hence how contested the exercise is.
2. Connected components of the allocation on the true adjacency graph — pricing contiguity rather than assuming it.
3. Measurement error on  $A_z, B_z, M_z$  — since data uncertainty, not parameter uncertainty, sets which zips need adjudication.

## A Other solution concepts

Selecting each criterion directly over the fifty-one prefixes of the instance in Section 5:

| crit <sup>erion</sup> | $k$ | $g_a$   | $g_b$  | $g_a g_b$     | $\min(g)$ | $ g_a/G_a^{\max} - g_b/G_b^{\max} $ |
|-----------------------|-----|---------|--------|---------------|-----------|-------------------------------------|
| <b>Nash</b>           | 22  | 7.4423  | 7.1628 | <b>53.308</b> | 7.1628    | 0.0016                              |
| equal gain            | 21  | 7.2232  | 7.3765 | 53.282        | 7.2232    | 0.0286                              |
| egalitarian           | 21  | 7.2232  | 7.3765 | 53.282        | 7.2232    | 0.0286                              |
| Kalai–Smorodinsky     | 22  | 7.4423  | 7.1628 | 53.308        | 7.1628    | 0.0016                              |
| equal opportunity     | 22  | 7.4423  | 7.1628 | 53.308        | 7.1628    | 0.0016                              |
| utilitarian           | 46  | 12.6330 | 2.0395 | 25.765        | 2.0395    | 0.7204                              |

Nash, Kalai–Smorodinsky and equal opportunity coincide here; equal gain and the egalitarian criterion agree with each other but differ from Nash by one zip. This is the reverse pairing from the pre-merger baseline, where Nash coincided with equal gain and egalitarian instead — a reminder that which criteria happen to agree is a property of the baseline, not a robust structural fact. Unlike the pre-merger baseline,  $G_a^{\max} = 14.604$  now exceeds  $G_b^{\max} = 14.100$  in the natural direction:  $a$  brought the bigger book ( $S_a = 3.0$  against  $S_b = 1.8$ ) and, at  $d = (0, 0)$ , also has the bigger attainable maximum, so Kalai–Smorodinsky no longer has to award more absolute gain to the smaller-book wholesaler to reach its target fraction — which is presumably why it now coincides with Nash rather than fighting it. The utilitarian row shows what fairness buys: unconstrained,  $a$  takes 46 zips and leaves  $b$  a gain of 2.04.

**Equalisation can destroy value.** Minimising a difference is not a monotone objective, and the unconstrained equalising MILP illustrates the hazard. At  $d = (0, 0)$ , minimising the KS gap over *all* subsets reaches a gap of  $5 \times 10^{-8}$  — essentially perfect equality — but at gains (7.323, 7.070) and product 51.77: 98.1% of attainable welfare against 99.5% for the Nash prefix, and Pareto-dominated by the Nash allocation (7.371, 7.232), so *both* wholesalers pay for the equality. (Under the discarded pre-merger baseline the same experiment was more dramatic — a gap of 0.000161 at 82.1% of welfare — the hazard shrinks at  $d = 0$  but does not vanish.) The remedy, if an equalising criterion must be used, is a welfare floor  $g_a + g_b \geq \kappa \max_S(g_a + g_b)$ , whose right-hand side is available in closed form because the utilitarian optimum is the prefix  $\{z : u_a(z) > u_b(z)\}$ . Nash needs no such guard.

## B The prefix heuristic

Section 3 solves (9) exactly by outer approximation. This appendix records a fast, solver-free alternative — a sanity check and a preview, not a substitute — and the extent to which it can be



trusted.

## B.1 Definition, and exactness for the utilitarian criterion only

Define the utility ratio

$$r_z = \frac{u_a(z)}{u_b(z)}, \quad (25)$$

the rate at which utility transfers between the wholesalers when zip  $z$  changes hands.

**Proposition 1** (Prefix heuristic). *Order the zips so that  $r_{(1)} \geq \dots \geq r_{(n)}$  and write  $P_k = \{(1), \dots, (k)\}$ . Then the utilitarian optimum is exactly  $P_k$  with  $k = |\{z : r_z > 1\}|$ , and  $\arg \max_k g_a[k]g_b[k]$  is a fast approximation to the Nash optimum.*

The prefix family has a second characterisation worth recording: allowing a single zip to be split fractionally, the optimal fractional allocation of two-agent maximum Nash welfare is exactly a ratio threshold — the discrete optimal-transport solution — so the *fractional* threshold value is an upper bound on the free Nash value, the best integral prefix sits within one zip’s contribution of it, and contiguity only restricts the feasible set further below both.

The utilitarian claim is exact: the objective is  $\sum_{z \in S} (u_a(z) - u_b(z)) + \text{const}$ , maximised by taking every zip with  $u_a > u_b$ . The Nash claim is not. The usual argument — at an optimum every included zip satisfies  $u_a(z)/g_a \geq u_b(z)/g_b$ , a threshold on  $r_z$  — is *marginal*, treating  $g_a, g_b$  as fixed while a zip moves. Discretely they jump by a finite amount, so the threshold characterisation is only asymptotically valid.

**The prefix rule is an approximation, and it is wrong about half the time.** Against exhaustive enumeration over 200 random instances per size:

| $n$ | prefix optimal | mean shortfall | worst shortfall |
|-----|----------------|----------------|-----------------|
| 6   | 40.0%          | 3.91%          | 53.9%           |
| 10  | 47.5%          | 1.17%          | 13.9%           |
| 14  | 45.0%          | 0.57%          | 10.7%           |

| $n$ | shortfall vs. local optimum |
|-----|-----------------------------|
| 20  | 0.16%                       |
| 50  | 0.02%                       |
| 100 | 0.002%                      |
| 400 | 0.0009%                     |

The error vanishes as the discretisation refines, because the jumps shrink relative to the gains. At realistic sizes it is far below the uncertainty in  $\theta$  or in the sales data — but it is not zero, and the rule should be described as a heuristic, never substituted for Section 3’s exact method where a certificate matters.



## B.2 Where it fails on other criteria

Proposition 1 covers Nash and the utilitarian criterion. It does not extend to criteria that minimise a *distance*, for which an off-frontier subset can hit the target more precisely. Over 400,000 random subsets of the worked instance:

| criterion                     | best prefix | best random subset |                 |
|-------------------------------|-------------|--------------------|-----------------|
| utilitarian $\max g_a + g_b$  | 14.673      | 14.630             | prefix holds    |
| Nash $\max g_a g_b$           | 53.308      | 53.101             | prefix holds    |
| egalitarian $\max \min(g)$    | 7.223       | 7.273              | <b>violated</b> |
| equal gain $\min  g_a - g_b $ | 0.1533      | 0.0000             | <b>violated</b> |
| Kalai–Smorodinsky             | 0.00160     | 0.00000            | <b>violated</b> |

The prefix table therefore remains exact for Nash, and is a fast restriction for the others — one with the merit of staying on a family of allocations ordered by exchange rate.

## C The mixture-quantile shortcut

**Open item: this appendix still assumes the discarded  $d = (S_a, S_b)$  baseline.** The derivation below collects  $g_a = g_b$  under the pre-merger convention, so  $q$ ’s numerator carries  $S_a, S_b$  terms that do not belong to (7) at  $d = (0, 0)$ . At the zero baseline the equal-gain condition is simply  $\sum_{z \in S} u_a(z) = \sum_{z \notin S} u_b(z)$ , i.e.  $\Phi(S) = q$  with  $S_a = S_b = 0$  substituted into  $q$  below — a strictly simpler target, not a harder one — but the closed-form  $K_a, K_b, K_M, q_{KS}$  for Kalai–Smorodinsky have not been re-derived at  $d = 0$ , and `verify_algebra.py` still certifies only the superseded baseline (HANDOFF.md trap list). Left as pending rather than silently repatched with unverified numbers.

For the *equalisation* criteria the continuum analysis yields an appealing shortcut. Collecting (7) under  $g_a = g_b$  gives

$$\Phi(S) = q, \quad \Phi(S) = \frac{\Lambda \sum_{z \in S} (A_z + B_z) + 2\lambda \sum_{z \in S} M_z}{D}, \quad q = \frac{S_a(1 + c_2) - \lambda S_b + \lambda M}{D}, \quad (26)$$

with  $\Lambda = c_1 + c_2$  and  $D = \Lambda(S_a + S_b) + 2\lambda M$ : accumulate mixture mass along the ratio order and cut where it crosses  $q$ . Kalai–Smorodinsky takes the same form with weights  $K_a = S_a(G_b^{\max} c_1 + G_a^{\max} c_2)$ ,  $K_b = S_b(G_b^{\max} c_2 + G_a^{\max} c_1)$ ,  $K_M = \lambda M(G_a^{\max} + G_b^{\max})$  and level  $q_{KS} = (G_a^{\max} G_b^{\max} + G_b^{\max} S_a)/K$ .

On a continuum this is exact. On discrete data it is off by one zip:  $q$  is a continuum target while the cumulative mass *jumps*, so truncation need not give the nearer prefix. With  $q_{KS} = 0.523326$  the crossing falls between  $k = 24$  (mass 0.517624) and  $k = 25$  (0.535206). On the equal-gain criterion the same truncation yields a gap of 0.449 where  $k = 25$  gives 0.055.

The mixture form remains valuable as *theory* — it makes  $q$  interpretable, yields closed-form comparative statics, and reveals that  $\lambda$  governs the weight on opportunity (Section 9) — but it is the wrong algorithm for discrete data, and it does not arise for Nash at all.

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